

Week 2: Data Basics

Day 3: Practical Activity & Documentation

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Abstract

Welcome to Day 3 of Week 2! Today, we transition from learning data cleaning techniques to executing a complete, documented data cleaning pipeline. In professional AI/ML projects, cleaning the data is only half the battle; documenting your methodology is equally critical. This session culminates in a guided practical activity where you will clean a dataset and build the documentation required for reproducible Machine Learning.

- **Documentation Principles:** Understanding reproducibility, transparency, and auditability in AI.
- **Key Artifacts:** Creating a "Data Dictionary" and a structured "Cleaning Log".
- **Best Practices:** Managing raw data, version control, and metadata.
- **Capstone Activity:** Independently cleaning a "Customer Churn" dataset and documenting every step.
- **Review & Next Steps:** Peer review, submission, and a preview of Week 3 (Principles of ML).

Session Objectives

By the end of today's session, learners will be able to:

- Explain why documentation is a non-negotiable requirement in professional AI/ML projects.
- Construct a comprehensive "Data Dictionary" for a given dataset.
- Maintain a rigorous "Cleaning Log" using the Problem-Tool-Solution framework.
- Apply best practices for raw data preservation and dataset versioning.
- Independently execute a full data cleaning pipeline on a sample dataset.
- Evaluate peer work and prepare cleaned assets for Machine Learning model training.

Part 1: The Importance of Documentation

"Execution and Documentation – The Data Cleaning Pipeline"

What is Documentation in AI/ML?

- In software engineering, documentation explains how code works. In AI/ML, documentation explains **how the data was prepared**.
- It is the "recipe" that bridges the gap between raw, messy information and a trustworthy, high-performing AI model.
- Without documentation, a cleaned dataset is just a "black box" – you know it's clean, but you don't know *how* it got that way.
- Documentation transforms a one-time data task into a **repeatable, auditable, and professional pipeline**.

The Core Philosophy

"If it isn't documented, it didn't happen." In AI, undocumented data cleaning is a massive liability.

Pillar 1: Reproducibility

- **Definition:** The ability for another data scientist (or yourself six months from now) to start with the exact same raw dataset and arrive at the exact same cleaned dataset by following your instructions.
- **Why it matters:** AI models are sensitive. If the data cleaning steps change slightly (e.g., using the Mean instead of the Median for imputation), the model's predictions will change.
- **The Test:** If you hand your Cleaning Log to a colleague, could they reproduce your final CSV without asking you a single question?

Pillar 2: Transparency & Trust

- Business stakeholders and end-users must trust the AI system.
- If an AI model predicts a high risk of customer churn, the business needs to know the data feeding that prediction wasn't manipulated or poorly handled.
- Transparent documentation shows exactly how missing values were filled, how outliers were treated, and how categories were standardized.
- **Result:** Stakeholders trust the model because they trust the data pipeline.

Pillar 3: Auditability & Compliance

- In regulated industries (Finance, Healthcare, Telecommunications), AI decisions are subject to legal and ethical audits.
- Auditors will ask: "How did you handle missing customer income data? Did you exclude certain demographics during cleaning?"
- A robust Cleaning Log provides the evidence required to prove that data was handled ethically, securely, and according to company policy.
- **Without documentation, compliance is impossible to prove.**

The Cost of Poor Documentation

What happens when we skip documentation?

- **Wasted Time:** Re-doing data cleaning from scratch because the original logic was forgotten.
- **Duplicated Effort:** Multiple team members cleaning the same dataset in different, conflicting ways.
- **Model Failure:** Deploying a model that fails in production because the training data was cleaned differently than the live data.
- **Loss of Credibility:** Inability to explain or defend AI-driven business decisions to leadership.

Key Documentation Artifacts

To achieve reproducibility, transparency, and auditability, every ML project requires two foundational documents:

1. The Data Dictionary

- A glossary of the dataset.
- Explains what each column means.
- Defines data types and allowed values.

2. The Cleaning Log

- A chronological record of changes.
- Details every manipulation applied to the raw data.
- Justifies the strategies used.

The Data Dictionary: Purpose

- **Context:** Column names like "Churn", "Tenure", or "MRR" might be obvious to the business team, but meaningless to a new data scientist or an ML algorithm.
- **Standardization:** It ensures everyone on the team agrees on what a variable represents.
- **ML Preparation:** It helps identify which columns are categorical (need encoding) and which are numerical (need scaling) before feeding data into a model.

Rule of Thumb

A Data Dictionary should be created *before* or *during* the cleaning process, not as an afterthought.

The Data Dictionary: Example

Column Name	Data Type	Description	Example Value
CustomerID	Text	Unique identifier for the client.	CUST-1042
Tenure	Numerical	Number of months the customer has been active.	24
Contract	Categorical	Type of contract (Month, 1-Year, 2-Year).	1-Year
Churn	Binary	Did the customer leave? (Yes/No).	Yes

This simple table eliminates ambiguity and serves as the blueprint for the ML model's feature engineering.

The Cleaning Log: Purpose

- While the Data Dictionary describes the *structure* of the data, the Cleaning Log describes the *history* of the data.
- It is a step-by-step narrative of every transformation applied to the raw dataset.
- It answers three critical questions for every change made:
  1. What was the **Problem**?
  2. What **Tool** was used?
  3. What was the **Solution** applied?

The Cleaning Log: The Framework

The Problem-Tool-Solution Framework

- **Problem:** Clearly state the data quality issue (e.g., "15 missing values in the 'Age' column").
- **Tool:** Specify the software and function used (e.g., "Excel `MEDIAN()` function").
- **Solution:** Describe the exact action taken and the justification (e.g., "Imputed missing values with the median age of 34, as the data contained outliers that skewed the mean").

This framework ensures that your log is actionable and scientifically rigorous.

The Cleaning Log: Example

Problem Identified	Tool Used	Solution Applied
15 missing values in 'Tenure'	Excel `MEDIAN()`	Imputed with median value (24 months).
5 exact duplicate rows	`Data > Remove Duplicates`	Deleted exact matches based on CustomerID.
Inconsistent 'Churn' labels	`Find & Replace`	Standardized "yes", "Yes ", "YES" to "Yes".

Notice the Detail

The log doesn't just say "Cleaned the data." It specifies *what* was cleaned, *how*, and *why*.

Best Practice: The Golden Rule

The Golden Rule of Data Handling

NEVER overwrite your original raw data file!

- The raw CSV is your "source of truth." If you make a mistake during cleaning, or if a stakeholder questions your methodology, you must be able to revert to the raw data and start over.
- Always keep the raw file in a "Read-Only" state or a separate "Raw_Data" folder.
- Perform all cleaning operations on a **copy** of the raw file.

Best Practice: Version Control

- As you clean data, you may need to try different strategies (e.g., dropping missing values vs. imputing them).
- Use clear, sequential file naming conventions to track your progress.

Example File Structure

- `customer_churn_RAW.csv` (Never touch this)
- `customer_churn_cleaned_v1.csv` (First attempt)
- `customer_churn_cleaned_v2_imputed.csv` (Final version with imputation)
- `customer_churn_FINAL_for_ML.csv` (Ready for Week 3)

Best Practice: Storing Metadata

- **Metadata** is "data about data." It provides context that isn't inside the dataset itself.
- Always document the following metadata alongside your Data Dictionary:
  - **Source:** Where did the data come from? (e.g., "Company CRM export, Oct 2026").
  - **Owner:** Who is responsible for this dataset?
  - **Date Created:** When was the raw file generated?
  - **Known Limitations:** Are there known biases or gaps in the data collection process?

Part 2: Guided Practical Activity

"Capstone for Week 2: Execution"

The Capstone Task: Overview

Your Mission

You will independently clean a provided raw dataset (`Customer_Churn_Raw.csv`) and create the complete documentation package required for a Machine Learning project.

- **Input:** A messy CSV file with duplicates, missing values, and formatting errors.
- **Tools:** Microsoft Excel or Google Sheets.
- **Output:**
  1. A clean, ML-ready CSV file.
  2. A Data Dictionary.
  3. A detailed Cleaning Log.

Meet the Dataset: Customer Churn

- **Context:** A telecommunications company wants to understand why customers are leaving (churning).
- **Dimensions:** Approximately 1,000 rows (customers) and 8 columns (features).
- **Target Variable:** `Churn` (Yes/No).
- **Known Issues (Do not peek!):** The dataset contains duplicate entries, missing values in the `Tenure` and `MonthlyCharges` columns, and inconsistent text formatting in the `Contract` column.

Let's begin the 5-step cleaning pipeline.

Step 1: Explore (Dimensions & Structure)

Action

- Open the raw CSV in your spreadsheet tool.
- Check the dimensions: Count the total rows and columns.
- Review the headers: Do they make sense? Are there any unnamed columns?
- Scroll through the first 20-30 rows to get a "feel" for the data.

Documentation Step

Record the initial row and column count in your Cleaning Log. This is your baseline.

Step 1: Explore (Spotting Obvious Errors)

- Look for obvious visual anomalies:
  - Are there blank rows at the bottom of the dataset?
  - Are there columns where all values are the same?
  - Do numerical columns contain text (e.g., "$50.00" instead of "50")?
- Use **Sorting** to push blanks or weird characters to the top/bottom.

Exploration is about forming hypotheses about what needs to be cleaned.

Step 2: Clean (Removing Duplicates)

Action

- Identify the primary key (e.g., `CustomerID`).
- Use the `Remove Duplicates` tool.
- Ensure you select the correct columns to define uniqueness.

Documentation Step

Log the exact number of duplicates removed. (e.g., "Removed 12 duplicate rows based on CustomerID").

Step 2: Clean (Handling Missing Values Strategy)

Before filling blanks, decide on your strategy:

- **Drop the Row:** If the missing value is in the Target column (`Churn`), or if the row is missing too much critical info.
- **Impute with Mean/Median:** For numerical columns like `Tenure` or `MonthlyCharges`. (Use Median if there are outliers).
- **Impute with Mode:** For categorical columns like `Contract`.
- **Fill with "Unknown":** If the absence of data is itself a meaningful signal.

Step 2: Clean (Handling Missing Values Execution)

Action

1. Calculate your chosen statistic (e.g., Median of `Tenure`).
2. Filter the column to show only "Blanks".
3. Fill the blank cells with your calculated value.
4. Clear the filter and verify no blanks remain.

Documentation Step

Log the strategy used and the exact value inserted. (e.g., "Imputed 45 missing 'Tenure' values with the median of 24").

Step 2: Clean (Fixing Formatting)

- **Trim Spaces:** Use the `TRIM()` function to remove accidental leading/trailing spaces (e.g., "Yes " becomes "Yes").
- **Standardize Categories:** Use `Find & Replace` or `PROPER()` to fix inconsistencies (e.g., "month-to-month", "Month", "MTM" $\rightarrow$ "Month-to-Month").
- **Fix Data Types:** Ensure numerical columns are actually formatted as Numbers, not Text.

Consistent formatting is critical; otherwise, the ML model will treat "Yes" and "Yes " as two completely different categories.

Step 3: Filter & Sort (Organizing Logically)

Action

- Sort the dataset by the Target variable (`Churn`) to see the distribution of Yes vs. No.
- Sort numerical columns to verify that your imputation didn't introduce impossible values (e.g., negative tenure).
- Apply final filters to ensure no hidden rows remain.

While ML models don't strictly require sorted data, sorting helps *you* verify the final quality of the dataset before export.

Step 4: Document (Writing the Log)

Action

- Open your Word document or the "Notes" tab in your spreadsheet.
- Review every action you took today.
- Translate your actions into the **Problem-Tool-Solution** framework.
- Ensure the tone is objective and precise.

Self-Correction

If you realize you forgot to document a step, go back and add it now. The log must be a perfect reflection of the work done.

Step 4: Document (Drafting the Dictionary)

Action

- List every column in your final dataset.
- Define the data type (Numerical, Categorical, Binary, Date).
- Write a clear, business-friendly description of what the column represents.
- Note any specific transformations applied (e.g., "Original text categories were consolidated into 3 standard groups").

The Data Dictionary is the ultimate handover document for the Data Science team.

Step 5: Export (Saving the Final CSV)

Action

1. Ensure all filters are cleared (no hidden rows).
2. Delete any helper columns or calculation cells you used during cleaning.
3. Go to `File > Save As > CSV (Comma delimited)`.
4. Name the file clearly: `Customer_Churn_FINAL_for_ML.csv`.

Crucial Reminder

Do not save the final ML input as an `.xlsx` file. ML tools (like Python, AutoML, or Azure) require pure CSV format to parse the data correctly.

Part 3: Review, Q&A, and Submission

"Peer Review, Instructor Feedback, and Next Steps"

Peer Review Activity

Instructions

1. Pair up with the person sitting next to you.
2. Swap laptops (or share screens).
3. You have 10 minutes to review your partner's work.

- **Your Goal:** Act as the "Lead Data Scientist" auditing their work. Can you understand their Cleaning Log? Does their Data Dictionary make sense?

Peer Review: The Checklist

Use this checklist to evaluate your partner's submission:

- ☐ Is the original raw file completely untouched and intact?
- ☐ Does the Cleaning Log detail *every* single change made?
- ☐ Is the Data Dictionary accurate, complete, and easy to understand?
- ☐ Are the final CSV columns properly formatted (no mixed data types)?
- **The Ultimate Test:** Could you reproduce their exact cleaned file using only their log and the raw data?

Instructor Review: Common Mistakes (Part 1)

Based on previous cohorts, here are critical errors to avoid:

1. The "Overwrite" Disaster

Saving the cleaned data over the raw file. If you realize you made a mistake, you have lost your baseline. *Always use versioning.*

2. The "Vague Log"

Writing "Cleaned missing values" in the log. This is useless for reproducibility. *Always specify the strategy and the exact value used.*

Instructor Review: Common Mistakes (Part 2)

3. Hidden Data

Forgetting to clear filters before exporting the CSV. The final file will be missing hundreds of rows, and the ML model will be trained on a biased subset.

4. Inconsistent Categories

Leaving "Nairobi", "nairobi", and "NBO" as separate categories. The ML model will treat these as three distinct cities, fragmenting the data. *Always standardize text.*

Submission Guidelines

What to Upload

By the end of today's session, upload the following to the Mediacrest learning portal:

1. **The Final Cleaned Dataset:** `Customer_Churn_FINAL_for_ML.csv`
2. **The Documentation Package:** A single PDF or Word doc containing your Data Dictionary and Cleaning Log.

Deadline

Submissions close at 17:00 EAT today. Late submissions will not be reviewed before Week 3 begins.

Week 3 Preview: Principles of Machine Learning

Next Week: From Clean Data to Smart Models Now that you have mastered data cleaning, we will learn how machines actually "learn" from this clean data.

- **Supervised vs. Unsupervised Learning:** Teaching with answers vs. discovering hidden patterns.
- **Classification vs. Regression:** Predicting categories (Churn: Yes/No) vs. predicting numbers (Sales: KES 50,000).
- **Training vs. Testing Data:** Why we hide some data from the model during the learning process.

Week 3 Preview: The ML Pipeline

The Big Picture

1. **Data Collection** (Week 1 & 2)
2. **Data Cleaning** (Week 2 - You just did this!)
3. **Data Splitting** (Week 3 - Train/Test split)
4. **Model Training** (Week 3 & 4 - The AI learns)
5. **Evaluation** (Week 4 - How accurate is it?)

Bring your cleaned `Customer_Churn_FINAL_for_ML.csv` to next week's class. We will use it to train our first model!

Q&A and Wrap-up

Questions?

"Documentation is the bridge between raw data and trustworthy AI."

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